

例子

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1 Topology 拓扑

Example 1.1. Assume X is an infinite set with cofinite topology. Then it is easy to check that X is noetherian and the only irreducible closed subset of X is itself so that X is of dimension 0. While given any open covering $\mathcal{U}$ of X , we cannot refine it to $\mathcal{V}$ satisfying that there exists n_0 such that any $n_0 + 1$ open subsets in $\mathcal{V}$ have no common element. Conclude that X has covering dimension infinite.

2 Abstract Algebra 抽象代数

Example 2.1. Assume $A = F_2$, $B = F_3$, where F_2, F_3 are respectively the free group generated by 2 and 3 elements. Then there exists natural injective homomorphism from A to B

$$\begin{aligned} f : A &\longrightarrow B \\ a &\longmapsto a' \\ b &\longmapsto b' \end{aligned} \tag{1}$$

also injective homomorphism from B to A

$$\begin{aligned} g : B &\longrightarrow A \\ a' &\longmapsto a^2 \\ b' &\longmapsto b^2 \\ c' &\longmapsto ab \end{aligned} \tag{2}$$

Example 2.2. $\mathbb{Z}_p$ for a prime number is all abelian simple groups.

Example 2.3. For every finite field $\mathbb{F}$, $PSL_n(\mathbb{F}) = SL_n(\mathbb{F})/Z(SL_n(\mathbb{F}))$ is simple group when $n \geq 2$.

Example 2.4. A_5 is the smallest nonabelian simple group.

Example 2.5. For special linear group, we have

$$Aut(SL_n(\mathbb{Q})) \cong Aut(PGL_n(\mathbb{Q})) \cong \begin{cases} PGL_n(\mathbb{Q}) \rtimes \{1, \psi\} & n \geq 3 \\ PGL_n(\mathbb{Q}) & n = 2 \end{cases} \tag{3}$$

where ψ is a canonical automorphism of $GL_n(\mathbb{Q})$

$$\begin{aligned} \psi : GL_n(\mathbb{Q}) &\longrightarrow GL_n(\mathbb{Q}) \\ A &\longmapsto (A^{-1})^T \end{aligned} \tag{4}$$

Example 2.6. If $n \neq 6$, then $Aut(S_n) = Inn(S_n)$. When $n = 6$, there exists an automorphism

of S_6 which is not inner

$$\begin{aligned}
 \psi : S_6 &\longrightarrow S_6 \\
 (1\ 2) &\longmapsto (1\ 2)(3\ 4)(5\ 6) \\
 (2\ 3) &\longmapsto (1\ 4)(2\ 5)(3\ 6) \\
 (3\ 4) &\longmapsto (1\ 3)(2\ 4)(5\ 6) \\
 (4\ 5) &\longmapsto (1\ 2)(3\ 6)(4\ 5) \\
 (5\ 6) &\longmapsto (1\ 4)(2\ 3)(5\ 6)
 \end{aligned} \tag{5}$$

And we have $\text{Aut}(S_6) \cong S_6 \rtimes \{1, \psi\}$

Example 2.7. Here is an example which is solvable group but not nilpotent group.

$$G = \begin{pmatrix} \mathbb{C}^\times & \mathbb{C} & \mathbb{C} \\ 0 & \mathbb{C}^\times & \mathbb{C} \\ 0 & 0 & \mathbb{C}^\times \end{pmatrix} \supseteq G^{(1)} = \begin{pmatrix} 1 & \mathbb{C} & \mathbb{C} \\ 0 & 1 & \mathbb{C} \\ 0 & 0 & 1 \end{pmatrix} \supseteq G^{(2)} = \begin{pmatrix} 1 & 0 & \mathbb{C} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \supseteq G^{(3)} = I_3 \tag{6}$$

Example 2.8. Take $G_0 = D_8$, $G_1 = \langle r^2, s \rangle$, $G_2 = \langle s \rangle$ and $G_3 = \{1\}$. Then $G_0 \supseteq G_1 \supseteq G_2 \supseteq G_3$ is a composition series which cannot be obtained by refining some normal series.

3 Algebraic number theory 代数数论

Example 3.1. L/K are Galois extension of number fields. Then $k(\mathfrak{P})/k(\mathfrak{p})$ is extension of finite fields.

$$\begin{aligned}
 k(\mathfrak{p}) &\cong \mathbb{F}_q, \quad q = N(\mathfrak{p}) = \#k(\mathfrak{p}) \\
 k(\mathfrak{P}) &\cong \mathbb{F}_{q^f}, \quad f = [k(\mathfrak{P}) : k(\mathfrak{p})]
 \end{aligned} \tag{7}$$

Moreover, $\text{Gal}(k(\mathfrak{P})/k(\mathfrak{p})) \cong \mathbb{Z}/f\mathbb{Z}$ generated by Frobenius automorphism $\sigma_q : \mathbb{F}_{q^f} \longrightarrow \mathbb{F}_{q^f} \quad x \longmapsto x^q$ which is canonical and natural.

4 Commutative Algebra 交换代数

Example 4.1. Take $G = \mathbb{Z}[\frac{1}{p}]/\mathbb{Z}$. Then G is Artinian as $\mathbb{Z}$ -module but not noetherian.

Example 4.2. $\mathbb{Z}[\frac{1}{p}]$ and $k[x_1, x_2, \dots]$ are neither noetherian nor Artinian.

Example 4.3. A ring with only one prime ideal may not be noetherian. Let k be a field and $A = k[x_1, x_2, \dots]$, $I = (x_1, x_2^2, \dots, x_n^n, \dots)$. Then the ring A/I has only one prime ideal $(\overline{x_1}, \overline{x_2}, \dots)$ but not noetherian.

Remark 4.1. To show A/I has only one prime ideal, only need to show there is exactly one prime ideal of A containing I .

Example 4.4. The only valuations on $\mathbb{Q}$ are the p -adic valuations $|\cdot|_p$ for prime numbers p and the trivial valuation $|\cdot|_0$. Thus there is a one-to-one correspondence between $\text{Spec}(\mathbb{Z})$ and $\text{Spv}(\mathbb{Q})$.

Example 4.5. $\text{Spv}(\mathbb{Z}) = \text{Spv}(\mathbb{Q}) \cup \{|\cdot|_{0,p} \mid p \text{ is prime number}\}$, where $|\cdot|_{0,p}$ is the valuation induced by the trivial valuation on $\mathbb{F}_p$.

Example 4.6. Take $\mathbb{Z}$ -modules $A' = \mathbb{Z}$, $A = \mathbb{Z}$ and $A'' = \mathbb{Z}/2\mathbb{Z}$. There is an exact sequence $0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z}$, where A' and A are flat but A'' is not.

5 Algebraic Geometry 代数几何

5.1 Sheaves

Example 5.1 (Sheaf Hom Doesn't Commute with Taking Stalk). Take $X = \mathbb{R}$ as topological space with structure sheaf constant sheaf associated to $\mathbb{Z}$. Consider the constant sheaf associated to $\mathbb{Z}$ over $\mathbb{R} \setminus \{0\}$ and push forward it to X , denoted by $\mathcal{F}$. Then $\mathcal{F}_0 = 0$ while for all open neighbourhood $U \ni 0$, $\mathcal{F}|_U \neq 0$. Thus $\text{Hom}(\mathcal{F}_0, \mathcal{F}_0) = 0$ while $\text{Hom}(\mathcal{F}, \mathcal{F})_0 \neq 0$.

5.2 Affine schemes

Example 5.2. Take $A = k[x_1, x_2, \dots]$ and $\mathfrak{m} = (x_1, x_2, \dots)$. Then $D(\mathfrak{m}) = \cup_i D(x_i)$ is an open subset of $\text{Spec } A$ which is not quasi-compact.

Example 5.3. Take R to be a DVR. Assume $\mathfrak{m} = (\varpi)$ is the maximal ideal of R and $k = R/\mathfrak{m}$ is the residue field. Then $\text{Spec } R[x]$ has two maximal ideals of different height, which are $(1 - \varpi x)$ and (x, ϖ) .

Example 5.4. Take $X = \text{Spec}(k[x, y, z]/(z^2 - xy))$. Then X is normal and noetherian but not regular at $(\bar{x}, \bar{y}, \bar{z})$.

Example 5.5. Take $A = \mathbb{Z}[x]/(2x)$. Then $\text{Spec } A$ is connected but not irreducible with two different minimal prime ideals (2) and (x) . Moreover, if take $A = \mathbb{Z}[x]/(4x)$, then $\text{Spec } A$ is even not reduced.

5.3 Spectrum maps

Example 5.6. The morphism $\text{Spec } \mathbb{Z}_p \longrightarrow \text{Spec } \mathbb{Z}$ is not of finite type.

Example 5.7. Let k be a field and $A = k[x_1, x_2, \dots]/(x_1^2, x_2^2, \dots)$. Then we have that $A/\mathfrak{N} = k$ which is noetherian. Note that $\text{Spec } A = \text{Spec}(A/\mathfrak{N})$, we get an example that A is not noetherian but spectrum of it is noetherian.

Example 5.8. Take $A = k[x, y]/(x^2, xy)$ and $X = Y = \text{Spec } A$, where k is algebraically closed. Take $f = \text{id}_A^*$ and $g = \varphi^*$, where φ is given by $\varphi(x) = 0$ and $\varphi(y) = y$. Then these give a counterexample of Hartshorne exercise ii 4.2(a).

Example 5.9. Even though $\mathbb{A}_{\mathbb{C}}^1 \rightarrow \mathbb{C}$ is separated, of finite type and closed, it is not proper since the natural map $\mathbb{A}_{\mathbb{C}}^2 \rightarrow \mathbb{A}_{\mathbb{C}}^1$ is not closed mapping $V(xy - 1)$ to set of all closed points except (x) . This satisfies our exception because the complex manifold corresponding to $\mathbb{A}_{\mathbb{C}}^1$ is not compact.

Example 5.10. Let $f : \operatorname{Spec} \mathbb{C} \rightarrow \operatorname{Spec} \mathbb{R}$ be the canonical morphism. Consider the base change of f by itself, we get $f \times_{\operatorname{Spec} \mathbb{R}} f : \operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}) \rightarrow \operatorname{Spec} \mathbb{C}$. Note that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$, we get that $f \times_{\operatorname{Spec} \mathbb{R}} f$ is not an isomorphism. Hence bijection and injection are not stable under base change.

Example 5.11. Take $f : \operatorname{Spec} k[t, x, y]/(x^2, tx, xy) \rightarrow \operatorname{Spec} k[t]$. Then all fiber dimension would be 1. And the fiber at $t = 0$ is $\operatorname{Spec} k[x, y]/(x^2, xy)$ not purely dimensional. Hence for a morphism with constant fiber dimension can have not purely dimension fiber at some closed point.

Example 5.12. Take field k and $X = Y = \operatorname{Spec} k(x)$. Then $Y \rightarrow \operatorname{Spec} k$ is flat of relative dimension 0 but after base change, $X \times_{\operatorname{Spec} k} Y \rightarrow X$ is not of same relative dimension, since $\dim X = 0$ while $\dim(X \times_k Y) = \dim(k(x) \otimes_k k(x)) = 1$. Also this gives a counterexample that $X \times_k Y$ even when the projection maps are flat.

5.4 Examples with affine line with double origin

Example 5.13. Take X to be affine line and Y to be affine line with two origin. Take f, g to be two inclusions from X to Y respectively. Then these give a counterexample of Hartshorne exercise ii 4.2(b).

Example 5.14. Take X to be affine line and Y to be affine line with two origin. Then there is natural morphism $f : Y \rightarrow X$ and inclusion $i : X \hookrightarrow Y$. Note that f is not separated and i is a section of f , which is not closed immersion.

Example 5.15. Take $A = \mathbb{Z}[x]/(3x^2 + 2x + 1)$ and $B = \mathbb{Z}$. Then $f : \operatorname{Spec}(A) \rightarrow \operatorname{Spec}(B)$ is surjective, of finite type and quasi-finite but not integral, hence also not finite.

5.5 Sheaves of module

Example 5.16. Consider affine scheme $X = \operatorname{Spec}(\mathbb{Z})$. Take $M = \bigoplus_{p \text{ prime number}} \mathbb{Z}/p\mathbb{Z} \in \operatorname{Mod}_{\mathbb{Z}}$. Then the corresponding sheaf $\widetilde{M}$ is finitely generated locally but not a finitely generated $\mathcal{O}_X$ -module.

Example 5.17. $\mathcal{O}_{\mathbb{P}^1} \xrightarrow{\cdot x} \mathcal{O}_{\mathbb{P}^1}(1)$ is injective as sheaf map but not injective as bundle map.

5.6 Geometric properties

Example 5.18. Take $X = \operatorname{Spec}(\mathbb{R}[x, y]/(x^2 + y^2))$ over $\operatorname{Spec} \mathbb{R}$. Then X is integral but not geometrically irreducible thus not geometrically integral.

Example 5.19. Take $X = \operatorname{Spec} \mathbb{C}$ over $\operatorname{Spec} \mathbb{R}$. Then X is integral but not geometrically connected thus not geometrically integral.

Example 5.20. Take $X = \operatorname{Spec} k(u)$ over $\operatorname{Spec} k(u^p)$, where k has characteristic p . Then X is reduced but not geometrically reduced.

6 Homological Algebra 同调代数

Example 6.1. Take $\alpha = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. Then $\alpha\beta = \beta\alpha$. Take $G = \langle \alpha \rangle \times \langle \beta \rangle$ and field k of characteristic p . Consider k^3 as kG -module. Then $\dim_k(\text{Fix}^G(k^3)) = 2$ and $\dim_k(\text{Cofix}_G(k^3)) = 1$. Thus $\text{Fix}^G \not\cong \text{Cofix}_G$, which gives an example that left adjoint and right adjoint (of trivial functor) are not isomorphic.

7 Representation Theory 表示论

Example 7.1. Take $G_1 = D_8$ and G_2 is Quaternions. Then G_1 and G_2 have the same character tables.